Energy Conservation Diagnostics for EncounterNN and Zone-3 Fine-Stepping

Summary report for the SIMON fine-stepping comparison across dt = 0.06, 0.08, and 0.10 yr

This report consolidates the new energy diagnostics generated from the fine-stepping evaluator. The goal is to separate trajectory-level energy conservation from the one-time neural correction energy jump, then compare noNN, EncounterNN, and Zone-3 fine stepping across the three tested timesteps.

Source files: energy_diagnostics_dt0.06.txt, energy_diagnostics_dt0.08.txt, and energy_diagnostics_dt0.1.txt.

1. Methodology and Energy Metrics

For each saved state, the total mechanical energy is computed as kinetic plus gravitational potential energy:

```
E(t) = K(t) + U(t)
K(t) = 1/2 sum_i m_i |v_i(t)|^2
U(t) = - sum over i<j of G m_i m_j / r_ij(t)
```

The relative energy deviation is measured against the first state of the same trajectory:

```
relE(t) = |E(t) - E(0)| / (|E(0)| + 1e-30)
```

The reported trajectory-level diagnostics are: relE_final at the final sample, relE_max as the worst sampled deviation, relE_timeavg as the RMS-average deviation over the trajectory, and relE_p95 as the 95th percentile sampled deviation. These are computed for the global 100-year runs and for the isolated local windows.

EncounterNN has one additional diagnostic because it applies a direct state correction once. At the event, three energies are compared: E_entry at the Zone-3 entry, E_no_exit after the 0.5-year noNN window, and E_corr immediately after applying the NN residual. The event metrics are:

```
window_noNN = |E_no_exit - E_entry| / |E_entry|
NN_jump = |E_corr - E_no_exit| / |E_no_exit|
total_entry_to_corr = |E_corr - E_entry| / |E_entry|
```

The NN_jump value is the cleanest measurement of the direct energy shock from the neural correction. The total_entry_to_corr value preserves the old relE_corr meaning used in earlier output tables.

Reporting thresholds

Status	Range	Meaning
EXCELLENT	< 1e-5	Very clean
GOOD	1e-5 to < 1e-3	Safe
CAUTION	1e-3 to < 1e-2	Discuss but not a failure
HIGH	1e-2 to < 1e-1	Serious caution
UNSAFE	>= 1e-1	Energy-unsafe for winner selection

The status is reporting-only. It should not stop a run. For interpretation, methods with UNSAFE status should not be treated as energy-safe winners.

2. Summary Tables

2.1 Global 100-year energy diagnostics

dt	Method	relE_final	relE_max	relE_timeavg	relE_p95	NN_jump	Status
0.06	noNN	1.497e-04	3.773e-03	7.156e-04	1.597e-03	-	CAUTION
0.06	EncounterNN	9.410e-04	3.773e-03	8.763e-04	1.125e-03	9.954e-04	CAUTION
0.06	finer-k2	8.031e-04	3.773e-03	7.475e-04	9.802e-04	-	CAUTION
0.06	finer-k4	9.695e-04	3.773e-03	8.151e-04	9.863e-04	-	CAUTION
0.08	noNN	4.493e-04	1.861e-02	2.266e-03	4.407e-03	-	HIGH
0.08	EncounterNN	1.902e-04	1.861e-02	2.259e-03	4.186e-03	2.644e-04	HIGH
0.08	finer-k2	4.997e-03	1.861e-02	2.980e-03	5.090e-03	-	HIGH

dt	Method	relE_final	relE_max	relE_timeavg	relE_p95	NN_jump	Status
0.08	finer-k4	6.446e-03	1.861e-02	3.409e-03	6.510e-03	-	HIGH
0.10	noNN	5.368e-03	3.429e-02	5.479e-03	6.496e-03	-	HIGH
0.10	EncounterNN	2.285e-02	2.690e-02	1.937e-02	2.565e-02	1.827e-02	HIGH
0.10	finer-k2	5.759e-03	1.026e-01	1.299e-02	1.556e-02	-	UNSAFE
0.10	finer-k4	1.618e+00	1.634e+00	8.252e-01	1.618e+00	-	UNSAFE

2.2 Accepted EncounterNN event energy audit

dt	t_start	t_exit	window_noNN	NN_jump	total_entry_to_corr	Status
0.06	33.600	34.100	6.727e-06	9.954e-04	9.887e-04	GOOD
0.08	82.560	83.060	2.626e-04	2.644e-04	1.697e-06	GOOD
0.10	36.900	37.400	5.212e-03	1.827e-02	2.339e-02	HIGH

2.3 Isolated local energy diagnostics

dt	Method	relE_final	relE_max	relE_timeavg	relE_p95	NN_jump	Status
0.06	local-noNN	5.603e-05	1.778e-03	7.678e-04	1.723e-03	-	CAUTION
0.06	local-EncounterNN	9.207e-04	1.008e-03	7.209e-04	9.875e-04	9.954e-04	CAUTION
0.06	local-finer-k2	6.128e-04	1.114e-03	5.832e-04	1.044e-03	-	CAUTION
0.06	local-finer-k4	7.843e-04	9.437e-04	6.429e-04	8.705e-04	-	GOOD
0.08	local-noNN	3.321e-04	4.565e-03	1.142e-03	1.463e-03	-	CAUTION
0.08	local-EncounterNN	5.300e-04	4.565e-03	1.398e-03	2.105e-03	2.644e-04	CAUTION
0.08	local-finer-k2	4.416e-03	5.298e-03	3.988e-03	4.801e-03	-	CAUTION
0.08	local-finer-k4	5.873e-03	6.823e-03	5.403e-03	6.253e-03	-	CAUTION
0.10	local-noNN	4.743e-03	1.038e-02	4.741e-03	5.212e-03	-	HIGH
0.10	local-EncounterNN	2.505e-02	2.677e-02	2.345e-02	2.505e-02	1.827e-02	HIGH
0.10	local-finer-k2	2.391e-03	4.105e-03	2.413e-03	2.939e-03	-	CAUTION
0.10	local-finer-k4	5.036e-03	6.315e-03	4.739e-03	5.120e-03	-	CAUTION

2.4 Ensemble energy summary

dt	Method	mean relE_max	max relE_max	mean NN_jump	max NN_jump	N HIGH	N UNSAFE
0.06	noNN	3.773e-03	3.773e-03	-	-	0	0
0.06	EncounterNN	3.773e-03	3.773e-03	9.957e-04	9.980e-04	0	0
0.06	finer-k2	3.773e-03	3.773e-03	-	-	0	0
0.06	finer-k4	3.773e-03	3.773e-03	-	-	0	0
0.08	noNN	5.662e-02	1.532e-01	-	-	7	3
0.08	EncounterNN	6.309e-02	1.532e-01	1.527e-02	3.896e-02	7	3
0.08	finer-k2	5.655e-02	1.532e-01	-	-	7	3
0.08	finer-k4	5.655e-02	1.532e-01	-	-	7	3
0.10	noNN	8.526e-02	1.361e-01	-	-	4	6
0.10	EncounterNN	2.692e-02	2.719e-02	1.831e-02	1.865e-02	10	0
0.10	finer-k2	1.004e-01	1.373e-01	-	-	5	5
0.10	finer-k4	3.501e+00	3.420e+01	-	-	5	5

3. Visual Results

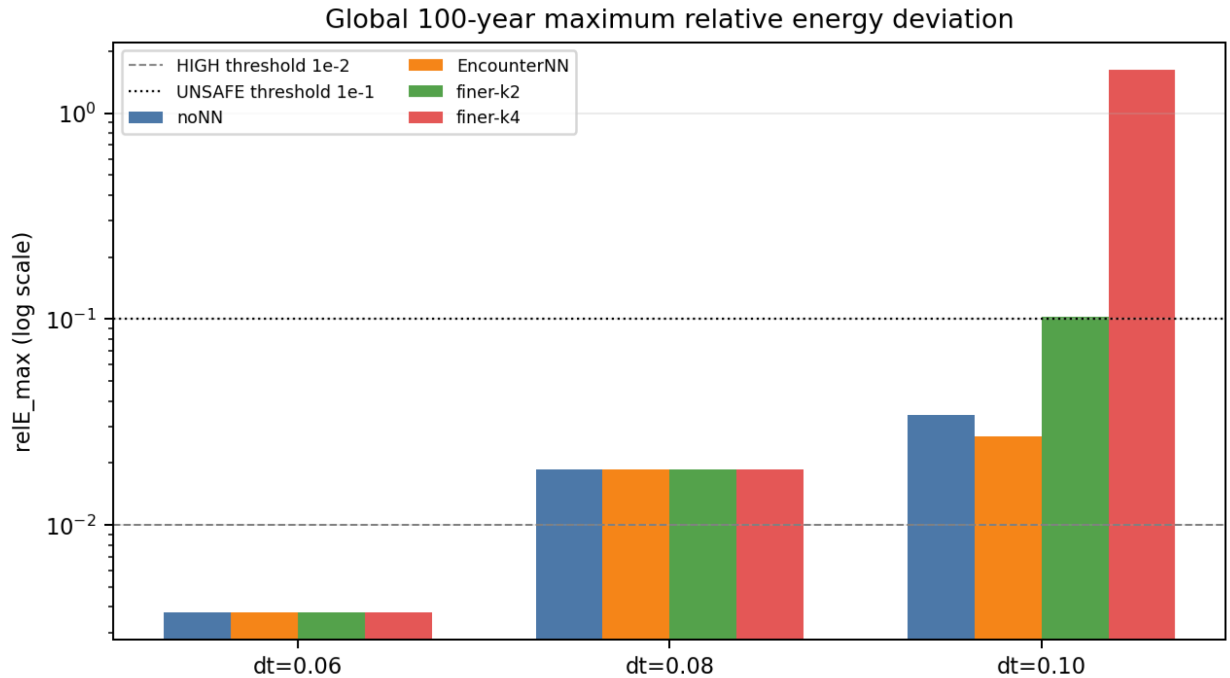

Figure 1. Global 100-year relE_max. dt=0.10 finer-k2 and finer-k4 cross the UNSAFE threshold; dt=0.08 is HIGH but not UNSAFE in the single-seed global run.

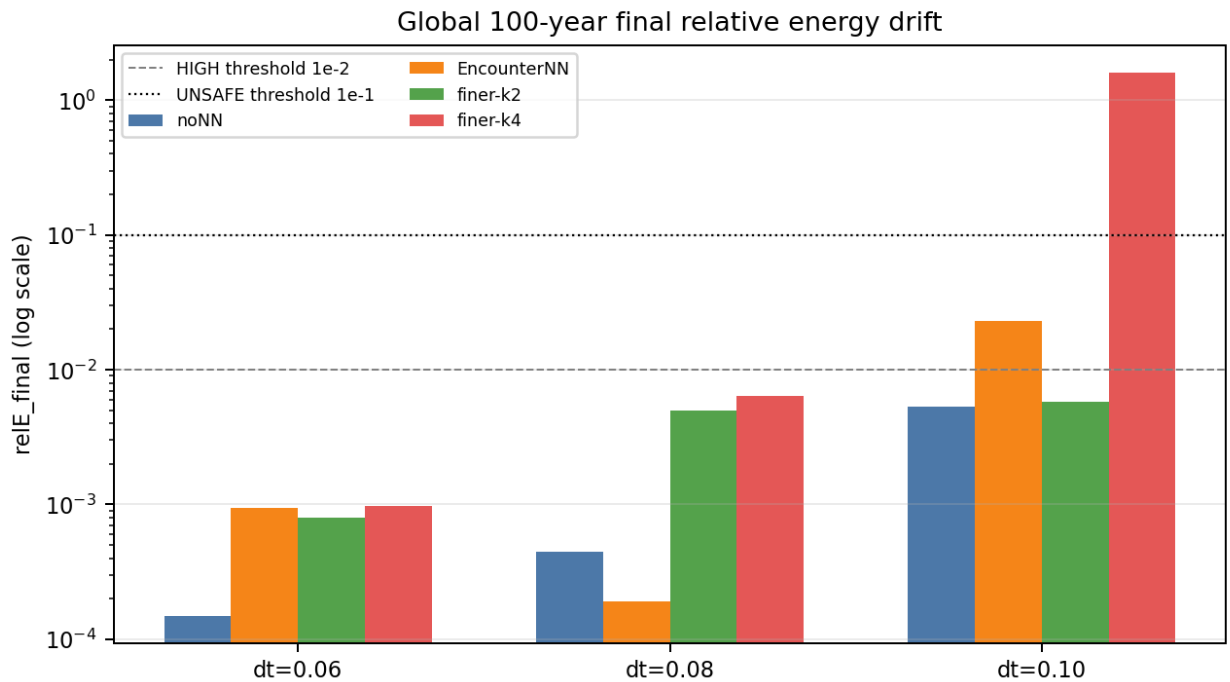

Figure 2. Global final energy drift. EncounterNN at dt=0.08 has the smallest final drift, while dt=0.10 finer-k4 is catastrophic.

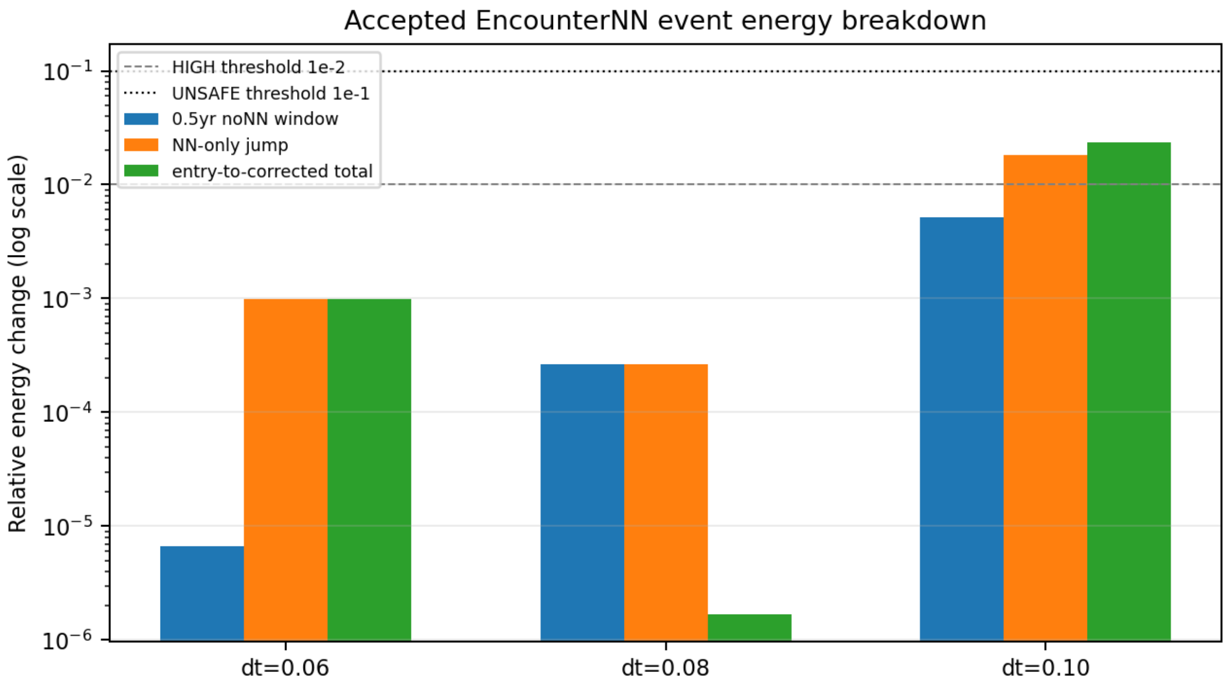

Figure 3. EncounterNN event energy breakdown. dt=0.08 has near cancellation between noNN-window drift and NN jump, producing a very small total_entry_to_corr.

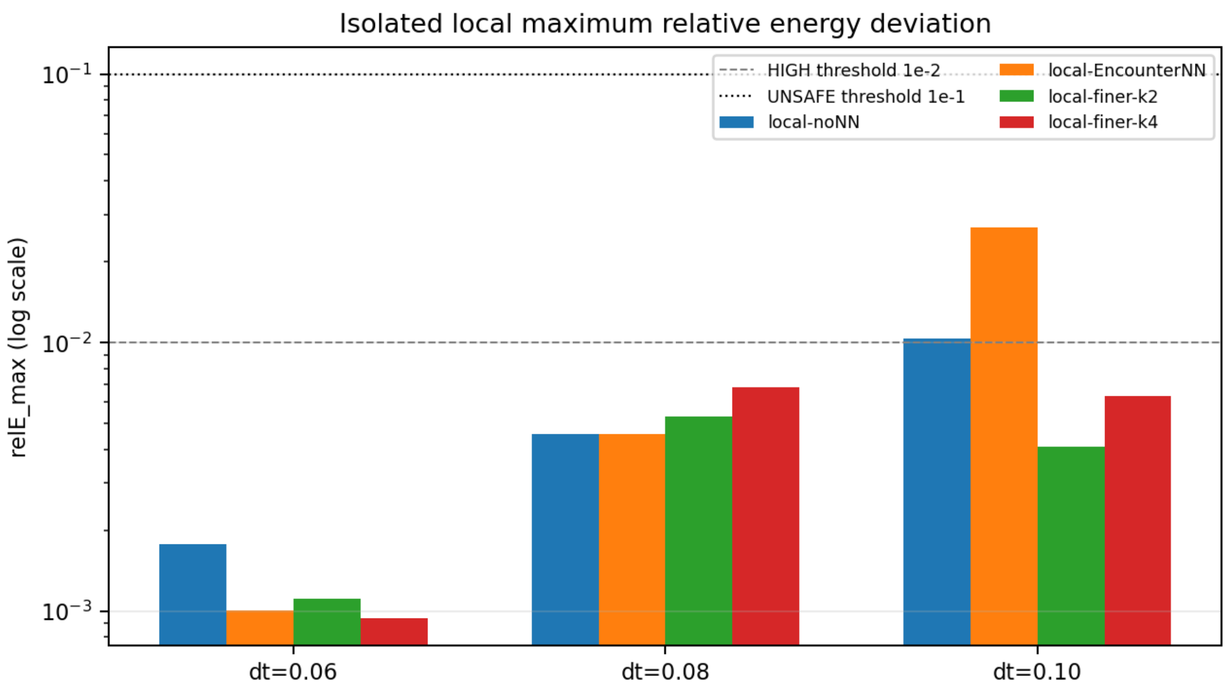

Figure 4. Isolated local relE_max. Local fine stepping is safer at dt=0.10, while EncounterNN has a larger local energy jump there.

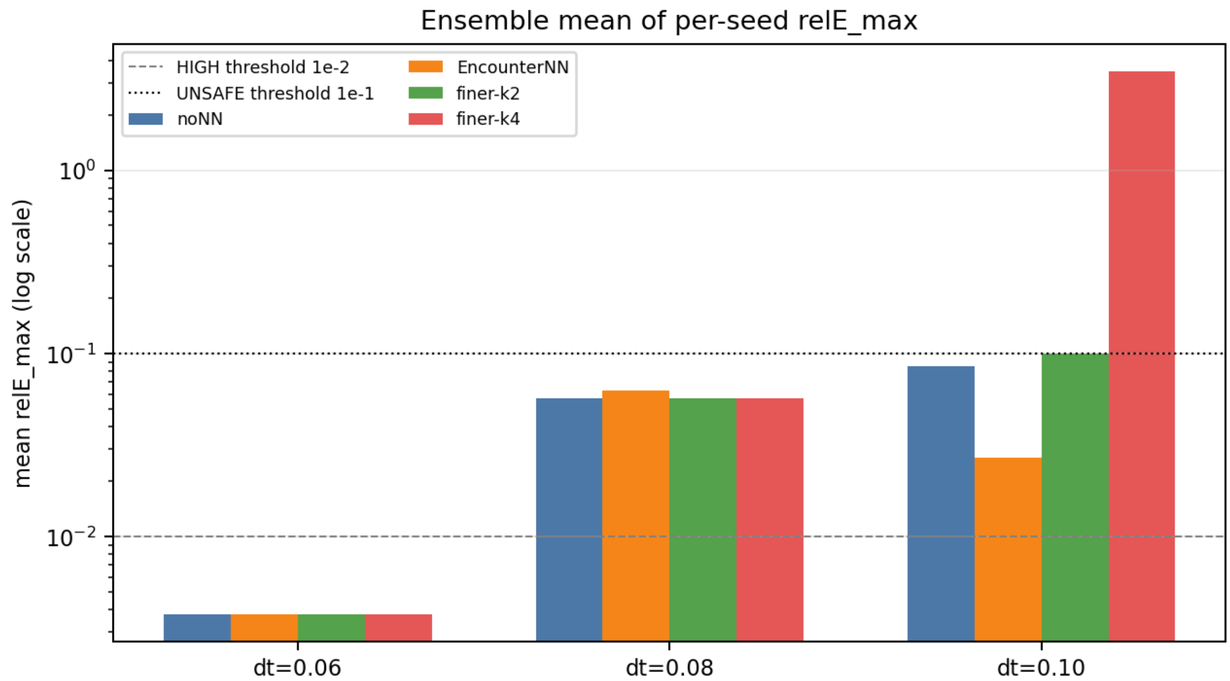

Figure 5. Ensemble mean relE_max across perturbed seeds.

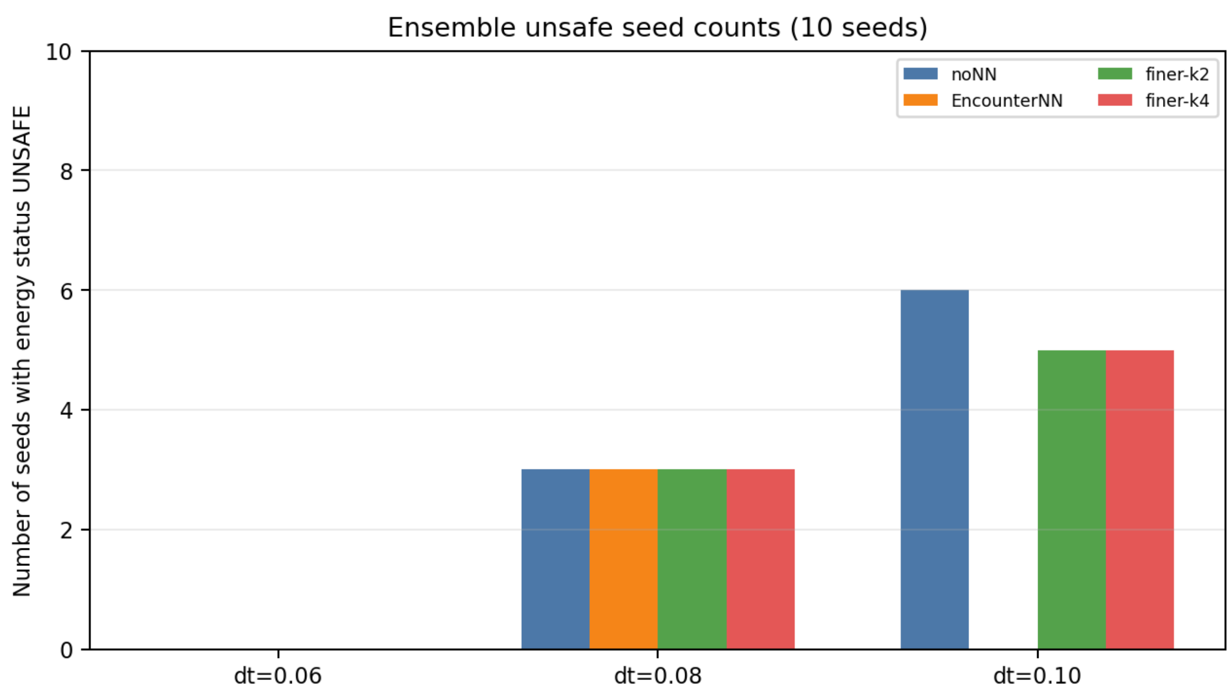

Figure 6. Ensemble unsafe seed counts. dt=0.10 fine stepping is unreliable; EncounterNN has HIGH energy status but no UNSAFE seeds in this ensemble.

4. Results and Discussion

At dt=0.06, all global methods remain below the UNSAFE threshold. The global relE_max is identical across methods at 3.773e-03, indicating that the maximum energy deviation is dominated by a shared part of the trajectory rather than the specific correction strategy. EncounterNN has an accepted event NN_jump of 9.954e-04 and total_entry_to_corr of 9.887e-04, both below 1e-3 or close to it. Energy therefore does not explain why EncounterNN loses global final-position accuracy at dt=0.06; the loss is more likely due to long post-encounter chaotic branch divergence.

At dt=0.08, the single-seed global relE_max is 1.861e-02 for all methods, which gives a HIGH status. However, EncounterNN has the smallest global relE_final at 1.902e-04 and the accepted event has total_entry_to_corr of only 1.697e-06. This is the strongest energy-conservation result for EncounterNN: its direct NN_jump is 2.644e-04, and it partly cancels the noNN-window drift of 2.626e-04. This supports dt=0.08 as the cleanest operating point for the neural correction.

At dt=0.10, the energy picture changes sharply. EncounterNN remains below the UNSAFE threshold in the single-seed global run, but its NN_jump rises to 1.827e-02 and its event status becomes HIGH. The rejected second event at t=71.0 yr has NN_jump 1.332e-01 and is correctly marked UNSAFE, showing that the max-one-correction rule and energy diagnostics are useful safeguards. Fine-stepping k2 becomes UNSAFE in the global run with relE_max 1.026e-01, and fine-stepping k4 becomes catastrophically unsafe with relE_final 1.618 and relE_max 1.634.

The ensemble results add a second layer. At dt=0.08, all methods have 3 UNSAFE seeds and 7 HIGH seeds, including noNN. This means some perturbed ICs are inherently difficult for this coarse hybrid setup and the unsafe energy behavior is not uniquely caused by EncounterNN. At dt=0.10, EncounterNN has no UNSAFE ensemble seeds, while noNN has 6, finer-k2 has 5, and finer-k4 has 5. However, EncounterNN still has HIGH status for all 10 seeds because its correction jump is around 1.8e-02. This is acceptable only as a cautionary result, not a clean operating point.

Overall, the energy diagnostics support three conclusions. First, EncounterNN is most energy-clean at dt=0.08, where the accepted correction is tiny in total energy effect and the global final drift is the smallest among methods. Second, dt=0.10 is not an energy-clean regime for aggressive fine stepping; k4 is clearly disqualified and k2 crosses the UNSAFE threshold in the global run. Third, energy conservation should be reported separately from positional accuracy: a method can be locally accurate but energetically questionable, or energy-safe but not positionally better after long chaotic divergence.

5. Recommended Paper Wording

For the energy conservation section, the safest concise wording is:

"Energy conservation was evaluated independently from trajectory accuracy using total mechanical energy E=K+U. For each method, we report relE_final, relE_max, relE_timeavg, and relE_p95 over the sampled trajectory. For EncounterNN, we additionally isolate the one-time neural correction energy jump, NN_jump, from the noNN drift during the 0.5-year local window. This separation is important because EncounterNN modifies the state directly once, whereas noNN and fine-stepping modify only the integration path."

"The cleanest EncounterNN energy result occurs at dt=0.08: the accepted event has NN_jump=2.644e-04 and total_entry_to_corr=1.697e-06, while the 100-year relE_final is 1.902e-04. At dt=0.10, the accepted correction produces a larger HIGH-status energy jump of 1.827e-02, and a second potential event is rejected with an UNSAFE jump of 1.332e-01. Fine-stepping k4 is energy-unsafe at dt=0.10. Therefore, dt=0.08 is the preferred operating point when accuracy and energy consistency are considered together."